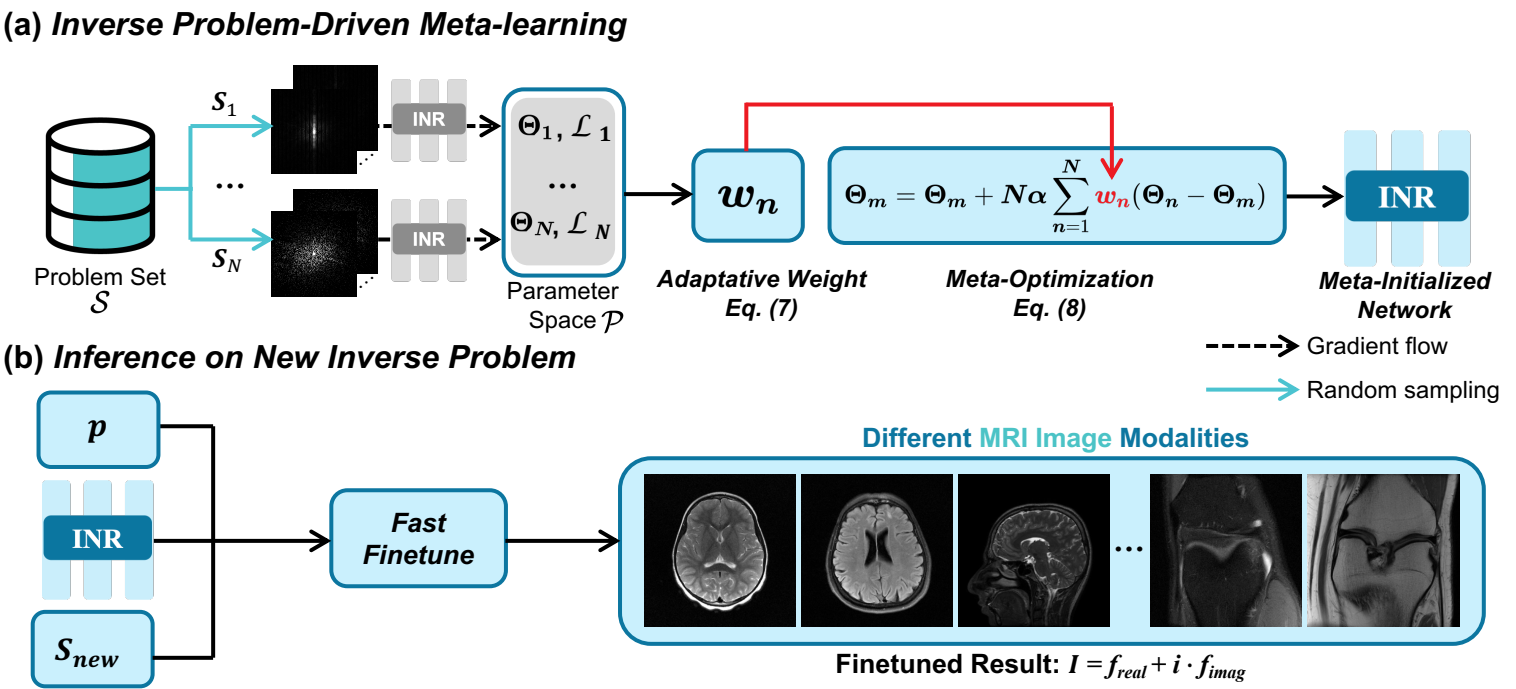
(a) Inverse Problem-Driven Meta-learning
Problem Set
$\mathcal{S}$
s_1
s_N
INR
INR
$\Theta_1, \mathcal{L}_1$
$\Theta_N, \mathcal{L}_N$
Parameter Space $\mathcal{P}$
w_n
Adaptative Weight Eq. (7)
$\Theta_m = \Theta_m + N\alpha \sum_{n=1}^{N} w_n(\Theta_n - \Theta_m)$
Meta-Optimization Eq. (8)
INR
Meta-Initialized Network
Gradient flow
Random sampling
(b) Inference on New Inverse Problem
p
INR
s_{new}
Fast Finetune
Different MRI Image Modalities
Finetuned Result: $I = f_{real} + i \cdot f_{imag}$